FULL NAME: ______________________________ **BLOCK:** ____________

nth Roots, Rational Exponents + Cylinder DOK-3 (Item 1A Prep)

Lesson 5-1 · Quick · Teacher Edition

OBJECTIVES

Math Objective	Relate nth roots to rational exponents, simplify radicals with variables, evaluate rational-exponent expressions, and use nth roots to solve geometric-modeling problems.
Language Objective	Explain why $(x^{(1)/(n)})^n = x$, using radical and rational-exponent notation interchangeably.
Essential Understanding	Rational exponents ARE radicals — $x^{(m)/(n)} = \sqrt[n]{x^m} = (\sqrt[n]{x})^m$. Choose whichever form makes the computation easier.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Fluency with nth roots and rational exponents; apply both to volume/radius modeling problems (Topic 5 assessment Item 1A prep).
Essential Question	When is rational-exponent form easier than radical form, and how do we rationalize denominators in cube-root expressions?
Materials	Student packet, pencil, calculator. DOK-3 Practice #50 (cylinder) is the lesson's capstone and rehearses the Topic 5 Assessment Item 1A move (derive variable from volume using rational exponents, then rationalize).

[PIN] PRE-FLIGHT — SINGLE-PERIOD COMPRESSED LESSON + DOK-3 CAPSTONE

This is a single-period quick treatment of Lesson 5-1. TWO examples are modeled in Launch (deviation from standard Klimsara one-example rule — content density requires it). Release promptly at minute 18.
Today's capstone is Practice #50 (cylindrical milk container). Students must (a) isolate r from $V = \pi r^2 h$ with $h = 2r$, giving $r = (V/(2\pi))^{(1)/(3)}$, (b) substitute into lateral surface formula $S = 2\pi rh = 4\pi r^2$, (c) revisit r to count stackable containers against the 20 ft cargo height. Same algebraic move as pyramid assessment Item 1A.
Pace: Try-It 1 (4 min) → Try-It 3 (4 min) → #28 (3 min) → #30 (3 min) → #37 (4 min) → #48 (4 min) → #50 (10 min). ∼32 min total Explore.
45-min Wed F variant: cut q37 + q48. Never cut #50.

Tag: bridge to nth roots launch

Do Now **DOK:** 1 **Minutes:** 5

Teacher does

- Hand out at door.
- Circulate 3 min.
- Cold-call 1 student for $y = x^2$ pair identification and root connection.

Students do

- Identify (x, y) pairs on $y = x^2$.
- Connect output y to the concept of nth root.

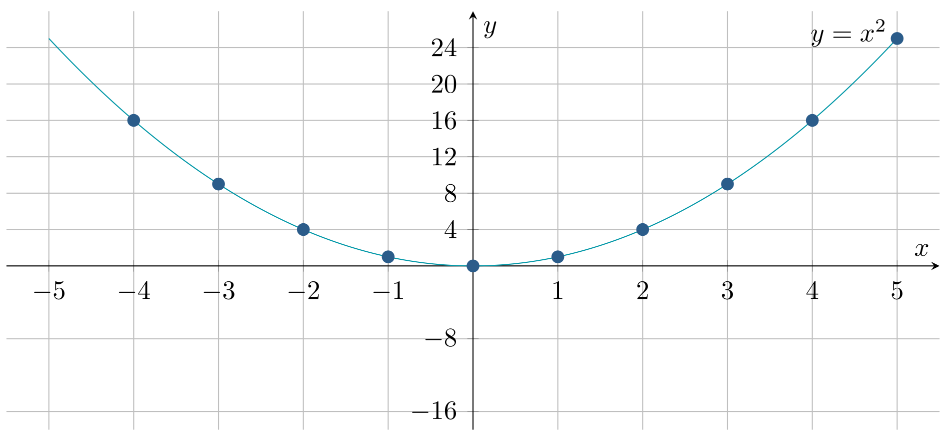
Questions to ask

- If $y = 16$, what are all real x values?
- What happens for negative y on $y = x^2$?

Adult role
Solo teacher: circulate silently. No hints.

Do Now — Explore & Reason: $y = x^2$ graph

EXPLORE & REASON
The graph shows $y = x^2$.
[GRAPH / TIKZ figure]



A. Find all possible values of x or y so that the point is on the graph.

(a) $(2, \_\_\_\_\_\_)$

(b) $(3, \_\_\_\_\_\_)$

(c) $(-3, \_\_\_\_\_\_)$

(d) $(5, \_\_\_\_\_\_)$

(e) $(\_\_\_\_\_\_, 4)$

(f) $(\_\_\_\_\_\_, -16)$

(g) $(\_\_\_\_\_\_, 7)$

(h) $(\_\_\_\_\_\_, 5)$

B. Communicate Precisely Write a precise set of instructions that show how to find an approximate value of $\sqrt{13}$ using the graph.

C. Draw a graph of $y = x^3$. Use the graph to approximate each value.

(a) $\sqrt[3]{5}$

(b) $\sqrt[3]{-5}$

(c) $\sqrt[3]{8}$

(d) A solution to $x^3 = 5$

(e) A solution to $x^3 = -5$

(f) A solution to $x^3 = 8$

[CHECK] ANSWER KEY

Answer key: A(a) 4, A(b) 9, A(c) 9, A(d) 25, A(e) $x = \pm 2$, A(f) no real value of x, A(g) $x = \pm\sqrt{7} \approx \pm 2.65$, A(h) $x = \pm\sqrt{5} \approx \pm 2.24$.
Part B sample instructions: Locate 13 on the y-axis, move horizontally to the graph of $y = x^2$, then move down to the x-axis; the positive x-value is $\sqrt{13} \approx 3.61$.
Part C: $\sqrt[3]{5} \approx 1.71$, $\sqrt[3]{-5} \approx -1.71$, $\sqrt[3]{8} = 2$, a solution to $x^3 = 5$ is ≈ 1.71, a solution to $x^3 = -5$ is ≈ -1.71, and a solution to $x^3 = 8$ is 2.

[TARGET] SENTENCE FRAME

"For $y = x^2$, the input $x =$ ___ gives output $y =$ ___. This means the square root of ___ is ___. There are ___ real values of x because ___."

LAUNCH — nth Roots + Rational Exponents (teacher models Ex 1 + Ex 3)

[BOOK] nth ROOTS ↔ RATIONAL EXPONENTS (keep printed on your page)

nth ROOTS ↔ RATIONAL EXPONENTS
1. $x^{(1)/(n)} = \sqrt[n]{x}$ (principal nth root).
2. $x^{(m)/(n)} = (\sqrt[n]{x})^m = \sqrt[n]{x^m}$.
3. To rationalize an nth-root denominator: multiply num/denom by $\sqrt[n]{x^{(n-1)}}$.
4. Negative nth roots: only odd n gives a real negative root; even n is undefined for negative radicands in Reals.

[WARN] COMPRESSED LESSON — TWO EXAMPLES IN LAUNCH

This is a single-period quick treatment. Teacher models BOTH Example 1 (real nth roots) AND Example 3 (evaluate rational exponents) during Launch.
Pay attention to the switch between radical form and rational-exponent form in Example 3 — that conversion is the core skill for the DOK-3 task.

Tag: teacher models Example 1 + Example 3 [COMPRESSED — 2 examples]

Launch **DOK:** 1–2 **Minutes:** 13

Teacher does

- Project Example 1. Think aloud: index n of root, principal root, even-index domain restriction.
- "For even n, the radicand must be non-negative. For odd n, any real works."
- Transition to Example 3 without stopping — announce: "Now watch the same concept in rational-exponent form."

- Project Example 3. Think aloud: $x^{(m)/(n)} \leftrightarrow (\sqrt[n]{x})^m$.
- Flag the Rules card — students copy into margin.
- 30-sec pause: "Turn to partner: rewrite your answer in the other notation."
- Do NOT solve Try-Its or Practice items — release at minute 18.

Students do

- Watch Example 1 silently. Note the even-index rule.
- Watch Example 3. Copy the $x^{(m)/(n)} \leftrightarrow$ radical conversion.
- During pause: convert partner's answer to the other notation.
- Copy Rules card into margin.

Questions to ask

- What is the index n in $\sqrt[4]{81}$? What principal root does it give?
- Why can't we take the square root of a negative number in Reals?
- Rewrite $8^{(2)/(3)}$ in radical form. Which form is easier to evaluate?
- If $x^{(1)/(n)} = \sqrt[n]{x}$, what is $(x^{(1)/(n)})^n$ equal to?

Adult role
Project both examples back-to-back. Release promptly at minute 18.

[MIC] TEACHER SCRIPT

1. "Watch me find a real nth root. The index tells me which root."
2. "Step 1: Identify the index n. Even index $\rightarrow$ radicand ≥ 0 in Reals."
3. "Step 2: Find the principal (non-negative) root."
4. "Now Example 3 — same idea, rational-exponent form."
5. "$x^{(m)/(n)} = (\sqrt[n]{x})^m$. I can go radical $\rightarrow$ exponent or exponent $\rightarrow$ radical — pick the easier path."
6. "Today's DOK-3 task uses this to derive a radius from a volume formula, then rationalize. That's the exact move on your Topic 5 Assessment."
7. Release to Explore.

EXPLORE — Think-Pair-Share (32 min)

Work with a partner. Move in order. Practice #50 (Cylinder Modeling) is the big one — plan ~10 min for it.

Tag: TPS + DOK-3 driver

Explore **DOK:** 2–3 **Minutes:** 32

Teacher does

- 5 laps across 32 min.
- Lap 1 (4 min): Try-It 1 — nth roots. Press pairs on index and principal root.
- Lap 2 (4 min): Try-It 3 — rational exponents. Watch for notation confusion.

- Lap 3 (3+3 min): #28 + #30. Press conversion between forms.
- Lap 4 (4+4 min): #37 + #48. Multi-step — confirm form choice.
- Lap 5 (10 min): [STAR] #50 Cylinder Modeling. Students set up $V = \pi r^2 h$ with $h = 2r$, solve for r in rational-exponent form, then rationalize.
- Do NOT give #50 answers. Pairs present argument at Share.

Students do

- 7 items in order: Try-It 1 → Try-It 3 → #28 → #30 → #37 → #48 → #50.
- For #50: BEFORE computing, write out $V = \pi r^2(2r) = 2\pi r^3$ and plan the rational-exponent isolation step.
- Justify rationalization step with sentence frame.

Questions to ask

- Try-It 1: what is the index? Is the radicand non-negative?
- #28: how do you simplify a variable inside an nth root with exponent?
- #30: convert to radical form first — which form is easier to evaluate?
- #37: do you simplify inside or outside the radical first?
- #48: what is $(x^{(1)/(3)})^3$? Use that to check your answer.
- #50: after substituting $h = 2r$, what formula do you get? Now isolate r using rational exponents.
- #50: to rationalize $\sqrt[3]{V/(2\pi)}$, what do you multiply numerator and denominator by?

Adult role
Solo teacher: 5 laps. On #50, press the substitution $h = 2r$ and the rationalization step — those are the assessment-critical moves.

Try It 1
Find the specified roots of each number.
a. real fourth roots of 81 **b.** real cube roots of 64

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): a. ± 3; b. 4

Try It 3
What is the value of each expression? Round to the nearest hundredth if necessary.
a. $-\left(16^{(3)/(4)}\right)$ **b.** $\sqrt[5]{3.5^4}$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key: a. -8; b. $3.5^{(4)/(5)} \approx 2.72$

Practice #28
Find the specified roots of each number.
The real square roots of 25.

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): ± 5

Practice #30
Rewrite each expression using a fractional exponent.
$\sqrt[6]{729}$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): $729^{\frac{1}{6}}$

Practice #37
Simplify each expression. Assume all variables are positive.
$\sqrt[6]{729a^{24}b^{18}}$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): $3a^4b^3$

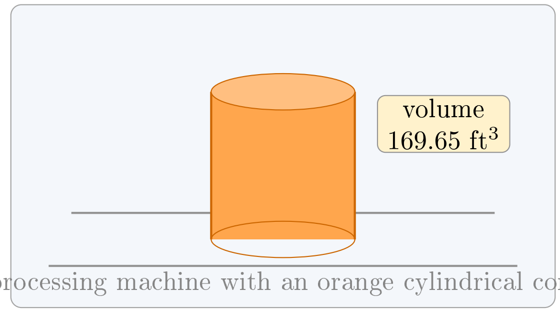
Practice #48
Determine if each expression is another way to write $b^{\frac{3}{4}}$. Select Yes or No.

Expression	Yes	No
a. $\sqrt[4]{b^3}$	☐	☐
b. $(b^3)^{\frac{1}{4}}$	☐	☐
c. $b^{\frac{4}{3}}$	☐	☐
d. $\sqrt[3]{b^4}$	☐	☐
e. $\dfrac{b^3}{b^4}$	☐	☐

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): a. Yes; b. Yes; c. No; d. No; e. No.

Practice #50 [STAR] DOK-3 CYLINDER MODELING
Performance Task A milk processing company uses cylindrical-shaped containers. The height of the container is equal to the diameter of the base.
[IMAGE: Milk processing machine with an orange cylindrical container; label reads "volume 169.65 ft^3."]



Milk processing machine with an orange cylindrical container

Part A How much material is needed to make the lateral surface area of the shipping container?
Part B The cargo hold of a ship is 20 ft high. What is the largest number of these shipping containers that could be stacked on top of each other inside the cargo hold?

[STAR] DOK-3 CYLINDER MODELING — Practice #50 (Topic 5 Assessment Item 1A prep)

Students must (a) isolate r from $V = \pi r^2 h$ with $h = 2r$, giving $r = (V/(2\pi))^{(1)/(3)}$, (b) substitute into lateral surface formula $S = 2\pi rh = 4\pi r^2$, (c) revisit r to count stackable containers against the 20 ft cargo height. Same algebraic move as pyramid assessment Item 1A.
Key criteria: (1) correct substitution $h = 2r$, (2) rational-exponent form for r, (3) rationalization shown with multiplication by $\sqrt[3]{(2\pi)^2}$.

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): Part A 36π ft^2. Part B 3 containers.
Teacher work: $V = \pi r^2 h$ with $h = 2r$ gives $169.65 = 2\pi r^3$, so $r = (169.65/(2\pi))^{(1)/(3)} \approx 3$ ft and $h = 6$ ft. Then $S = 2\pi rh = 4\pi r^2 = 36\pi$ ft^2, and $20/6 = 3$ whole containers can be stacked.

Tag: academic conversation + Topic 5 bridge

Share / Summary **DOK:** 2 **Minutes:** 5

Teacher does

- Cold-call 1 pair to present their cylinder solution using sentence frame.
- Class signals thumbs. Write $r = (V/(2\pi))^{(1)/(3)}$ on board.
- Topic 5 bridge: "This is the exact move on your Topic 5 Assessment Item 1A. You have now rehearsed it."

Students do

- Presenting pair reads their solution and cites the rationalization step.
- Rest of class signals and writes takeaway in margin.

Questions to ask

- Summarize: how did you convert $V = 2\pi r^3$ to $r = (V/(2\pi))^{(1)/(3)}$?
- Why did you need to rationalize? What did you multiply by?

Adult role
Cold-call a pair that struggled on #50 — hold them accountable to the rationalization step.

Sentence frames:
Pair share (Cylinder): "I isolated r from $V = \pi r^2 h$ by substituting $h = 2r$, giving $V =$ ___. Solving for r, I got $r =$ ___. In rational-exponent form, $r = (V/(2\pi))^{(1)/(3)}$. To rationalize I multiplied by ___ to get ___."
Whole class: one pair reads their cylinder solution. Class signals thumbs up/sideways.

[BRIDGE] BRIDGE TO TOPIC 5 ASSESSMENT

The cylinder move — isolate a variable, convert to rational-exponent form, rationalize — is exactly what the Topic 5 Assessment Item 1A asks. You have now rehearsed it.

Tag: summary recap (not graded)

Exit Ticket **DOK:** 1–2 **Minutes:** 3

Teacher does

- Collect at door.
- Scan stem 2 (rationalize). Plan re-teach if many students blank on the $\sqrt[n]{x^{(n-1)}}$ move.

Students do

- 3 sentence stems, honest.

Questions to ask

- Did students connect $x^{(m)/(n)}$ to radical form correctly?
- Did students explain the rationalization multiplier?

Adult role
Collect. No grade.

[CLIPBOARD] EXIT TICKET — Student Form

$x^{(m)/(n)}$ is the same as ___ in radical form.
To rationalize a cube-root denominator I ___.
One biggest thing I learned today: ____.

OPTIONAL REINFORCEMENT (not collected)

If you finish Explore early. #49 is an SAT/ACT cube-root item that extends rational-exponent fluency with large radicands.

Optional #1 (Savvas Practice #49 — SAT/ACT $\sqrt[3]{4{,}096x^{18}y^{30}}$)

SAT/ACT Which of the following is equivalent to $\sqrt[6]{4{,}096x^{18}y^{30}}$, where $x > 0$ and $y > 0$?

A. $682.7x^{15}y^{24}$

B. $4x^{1.6}y^{1.8}$

C. $4{,}096x^3y^5$

D. $4x^3y^5$

E. $682.7x^3y^5$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): D. $4x^3y^5$

[CLOCK] PERIOD A / PERIOD F — 65-MIN CLASS NOTES

Both sections should have 65 min for this lesson. Use the extra 10 min on Explore #50 (Cylinder Modeling) — the richer the rationalization argument, the better the Topic 5 Assessment prep payoff.
45-min Wed F variant: cut q37 + q48 from Explore. Never cut #50.
If finished early: hand out Practice #49 (SAT/ACT cube-root stretch).

[PUZZLE] MODIFICATIONS / SUPPORT FOR IEP STUDENTS

- Provide the nth Roots ↔ Rational Exponents Rules card as a sticker/cut-out.
- For #50, provide the setup: $V = \pi r^2 h$ with $h = 2r \to V = 2\pi r^3$. Students solve from that point.
- Accept verbal rationalization explanation in lieu of written steps.

[GLOBE] SUPPORTS FOR ELL STUDENTS

- Sentence frames on student page (don't skip).
- Word cards: "index" = the small number outside the radical; "rational exponent" = fraction exponent; "rationalalize" = remove the radical from the denominator.
- "Principal root" = the non-negative root (even index only).
- Pair ELL students with English-dominant partners for TPS.